SANDEEP VEERABATHINI
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PROFESSIONAL SUMMARY

- Data Engineer with 7 years of experience building scalable, cloud-native data platforms across automotive, insurance, banking, and capital-markets industries.
- Expert in AWS & Azure ecosystems with hands-on experience delivering high-performance batch, streaming, and ML-ready data pipelines using PySpark, Spark SQL, Databricks, EMR, AWS Glue, and Azure Data Factory.
- Specialized in AI/ML-ready data architectures, enabling feature computation, model training, inference, orchestration, monitoring, and automated retraining in production.
- Proven track record modernizing legacy on-prem systems and migrating high-volume vehicle, claims, underwriting, customer, and trading datasets into unified cloud data platforms.
- Strong experience building real-time ingestion and analytics pipelines using Kafka, Kinesis, Event Hubs, and Stream Analytics for fraud detection, telemetry, UX analytics, and risk monitoring use cases.
- Deep expertise designing canonical models, dimensional warehouses, semantic layers, and curated marts on Redshift, Synapse, and SQL platforms for enterprise reporting and ML feature engineering.
- Implemented enterprise-grade data quality, governance, and compliance frameworks (GDPR & financial regulations) with encryption, IAM/RBAC, credential vaulting, networking controls, auditing, and lineage monitoring.
- Skilled in workflow orchestration and DevOps for data using Airflow, Azure Data Factory, CI/CD pipelines, and IaC-driven deployments to ensure scalable, reliable, and auditable data operations.
- Strong cross-functional collaborator working directly with product, UX, claims, risk, trading, and data-science teams to convert complex requirements into actionable datasets, KPIs, and executive-ready insights.

Technical Skills

Big Data & Distributed Computing	Apache Spark, PySpark, Hadoop, MapReduce, HDFS, YARN, Hive, Impala, Sqoop, Oozie, Kafka/MSK, Kinesis, Apache Flume, Apache Airflow, Cloudera, HBase
Programming & Scripting	Python, SQL, T-SQL, PL/SQL, PySpark, Scala, Shell Scripting (Unix/Linux), PowerShell, C
Cloud Platforms	Amazon Web Services (AWS), Microsoft Azure
AWS Services	S3, EMR, Glue, Redshift, Lambda, EC2, DynamoDB, MSK (Kafka), RDS, VPC, CloudWatch, IAM, Kinesis, CloudFormation
Azure Services	Azure Data Factory, ADLS Gen2, Azure Databricks, Synapse Analytics, Azure SQL Database, Blob Storage, Azure Stream Analytics, Event Hubs, Azure Cosmos DB, Azure Monitor, Log Analytics, Key Vault, Azure DevOps
Databases & Data Warehousing	SQL Server, MySQL, Oracle, Teradata, Redshift, Snowflake, Azure SQL/Synapse, PostgreSQL
NoSQL Databases	Azure Cosmos DB, DynamoDB, HBase
ETL, Data Integration & BI Tools	Informatica, Talend, SSIS, Power BI, Tableau, OBIEE, SSRS
Version Control, CI/CD & DevOps	Git, Bitbucket, GitHub, Jenkins, Azure DevOps, SVN
Monitoring & Tracking Tools	Apache Airflow, CloudWatch, Azure Monitor, JIRA
Operating Systems	Linux, Unix, Windows, macOS
File Formats	JSON, CSV, Parquet, Avro, ORC

PROFESSIONAL EXPERIENCE

Client: Credit Acceptance, Atlanta, GA
Role: Data Engineer **Jan 2025-Present**

Responsibilities

- Managed end-to-end ETL and distributed data pipelines across Azure and hybrid environments, ensuring scalability, reliability, and high performance.
- Implemented optimized SQL query techniques, indexing strategies, and performance-tuning methods to improve data processing and retrieval.
- Performed complex data manipulation and transformation using SQL (including Dremio), PL/SQL, Python, and Perl.
- Integrated on-premise data sources (Oracle, MySQL, Cassandra) with cloud platforms (Azure Blob, Azure SQL DB) using Azure Data Factory, Apache NiFi, and REST APIs; loaded into Snowflake with required transformations.
- Designed scalable batch and streaming ingestion pipelines using Kafka, NiFi, and ADF to process large structured and semi-structured datasets.
- Built and maintained ETL/ELT workflows using Azure Databricks (Spark, PySpark, Scala) and Airflow to support analytical and ML workloads.
- Applied robust data quality and data cleansing techniques to ensure completeness, accuracy, and compliance through the pipeline lifecycle.
- Developed transformation logic using Spark SQL, Spark DataFrames, and Snowflake SQL to support consumable analytics-ready datasets.
- Collaborated closely with data architects, product owners, and compliance teams to deliver scalable and secure data-driven business solutions.
- Built and optimized data models and warehouse schemas leveraging Snowflake, Hive, and Azure Synapse for analytics and BI reporting.
- Supported hybrid cloud workloads with AWS Glue, EMR, Lambda and data movement across Azure Data Factory, AWS Step Functions, and GCP Workflows.
- Worked with ML teams to define and deliver feature stores in Snowflake and BigQuery for real-time and historical feature delivery to Vertex AI and SageMaker.
- Enabled LLM/GenAI initiatives by generating chunked document embeddings using PySpark and storing them in vector databases (Pinecone, Azure AI Search – POC level).
- Leveraged Apache Beam for unified batch and stream data processing in Azure Databricks and GCP Dataflow.
- Integrated AWS S3 and GCS with Snowflake external tables using secure stage objects and managed identities.
- Developed and optimized Azure Functions for data extraction and transformation from APIs, databases, file systems, and legacy Perl-based systems.
- Implemented CI/CD automation for data pipelines using Azure DevOps and Jenkins, improving deployment efficiency and reducing manual operations.
- Participated in Agile Scrum ceremonies, enabling transparent communication, sprint planning, and timely delivery in distributed teams.
- Provided production support including incident handling, performance troubleshooting, and SLA management for compliance and risk applications.
- Designed internal visualization dashboards using React and Power BI to enhance data transparency, monitoring, and business consumption.

Environment / Tools : Azure Databricks, Azure Data Factory, Synapse Analytics, Azure SQL DB, Azure Functions, Blob Storage, Snowflake, Hive, Spark, Python, Scala, PL/SQL, PySpark, NiFi, Apache Kafka, Apache Beam, Airflow, AWS Glue, EMR, Lambda, GCP Dataflow, DevOps (Azure DevOps, Jenkins, Git), Agile/Scrum, Power BI, React, Dremio SQL.

Client: Northern Illinois University, Illinois **May 2024 – Dec 2024**
Role: Data Analyst
Responsibilities:

- Developed robust SQL pipelines to extract and transform patient outcome, claims, and resource utilization data from EHR and billing systems, increasing data freshness by 50%.
- Built Power BI dashboards tracking admissions, discharge rates, and quality metrics, cutting manual report preparation time by 45%.
- Leveraged Python for advanced data cleansing, anomaly detection, and supporting better clinical and operational decision-making.
- Collaborated with clinicians, administrators, and finance teams to define KPIs and automate ETL workflows, improving

reporting consistency.
- Presented complex financial analytics in clear visuals and written briefs for executives and department heads, driving data-based process improvements.

Client: Siemens, India
Role: Data Engineer **Aug 2021 – July 2023**
Responsibilities:
- Built scalable batch and streaming ETL/ELT pipelines on AWS EMR to process high-volume data from APIs, databases, and file-based sources.
- Managed real-time ingestion using AWS Kinesis with downstream delivery to S3 and Redshift, improving data freshness and reducing latency.
- Optimized Redshift and Hive performance through partitioning, clustering, materialized views, and query tuning to reduce cost and execution time.
- Developed reusable PySpark modules and transformation frameworks to standardize data ingestion and accelerate onboarding of new data feeds.
- Automated end-to-end data pipelines using AWS Lambda and Step Functions, enabling reliable serverless orchestration and event-driven processing.
- Migrated large SQL workloads from Oracle to Redshift, improving scalability and query execution while minimizing downtime.
- Implemented IAM-based least-privilege access, governance controls, and secure data sharing across AWS environments.
- Provisioned cloud resources using AWS CloudFormation and integrated CI/CD deployments through GitHub Actions for pipeline automation.
- Created and maintained metadata assets including data dictionaries and ER diagrams using DBT and ERwin, improving lineage and data transparency.
- Documented data architecture, pipeline flows, and runbooks to support governance, troubleshooting, and production support readiness.

Environment : Python, Redshift, AWS Glue, AWS Kinesis, AWS EMR, Amazon S3, Cloud IAM, AWS Step Functions, PySpark, Oracle, DBT, ERwin, SQL Workbench, GitHub Actions, Vertex AI, Pub/Sub.

Client: Siemens, India
Role: Data Analyst **Jun 2018 – July 2021**
Responsibilities:
- Translated business requirements into technical specifications and design blueprints for healthcare, retail banking, and wealth management data solutions.
- Extracted, transformed, and analyzed large datasets using **SQL**, **Excel**, and **Python** to support data-driven decisions and regulatory compliance.
- Designed and automated recurring reports and dashboards using **Power BI**, **Excel VBA**, and **SQL ODBC**, reducing manual effort and improving KPI visibility.
- Developed **data validation**, **quality checks**, and **data reconciliation** frameworks to ensure accuracy and reliability in reporting.
- Built **forecasting models** and performed **trend analysis** to support performance tracking and predictive insights.
- Collaborated with **cross-functional teams** (data engineering, compliance, and business users) to define metrics, improve data governance, and align on reporting standards.
- Documented **data sources**, **ETL logic**, and **reporting workflows** in **JIRA** and **Visio** to maintain transparency and traceability.

- Supported ad hoc reporting requests and conducted **root cause analysis** on data discrepancies to improve reporting accuracy.
- Presented insights through interactive **Power BI** and **Tableau** dashboards, enabling leadership to make informed, data-backed decisions.
- Contributed to continuous improvement by recommending process automation, optimizing SQL queries, and enhancing dashboard performance.

EDUCATION

Master of Science – Management Information Systems |Northern Illinois University| USA *Aug 2023 – May 2025*
Bachelor of Technology – Computer Science | Indur Institute of Technology| India | *Aug 2015 – May 2019*